

Chapter 15 - Limits and Continuity

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#Limit of a Function

Limit of a function is the value of the function at which the function converges when the input value approaches a certain number.

$\lim_{x \rightarrow a} f(x) = L$ means that when x approaches to a then function $f(x)$ approaches to L .

#Indeterminate Forms

Indeterminate forms are the combinations of two functions, which cannot be determined solely (only) from the limits of individual functions. There are 7 indeterminate forms:

1. $\frac{0}{0}$
2. $\frac{\infty}{\infty}$
3. $0 \cdot \infty$
4. $\infty - \infty$
5. 1^∞
6. ∞^0
7. 0^0

#Algebraic Properties of Limit

1. $\lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$
2. $\lim_{x \rightarrow a} [f(x) g(x)] = \lim_{x \rightarrow a} f(x) \lim_{x \rightarrow a} g(x)$
3. $\lim_{x \rightarrow a} [k f(x)] = k \lim_{x \rightarrow a} f(x)$
4. $\lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$
5. $\lim_{x \rightarrow a} [f(x)]^n = [\lim_{x \rightarrow a} f(x)]^n$

#Standard Limits

Algebraic Limit

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}, (n \in \mathbb{Q})$$

Trigonometric Limits:

1. $\lim_{\theta \rightarrow 0} \sin \theta = 0$
2. $\lim_{\theta \rightarrow 0} \cos \theta = 1$
3. $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$
4. $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1$

Logarithmic and Exponential Limits:

1. $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$
2. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$ where $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$
3. $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a$

#Left Hand Limit and Right Hand Limit

1. **Left Hand Limit:** $\lim_{x \rightarrow a^-} f(x)$ means the limit as x approaches to a from the left/negative side.
2. **Right Hand Limit:** $\lim_{x \rightarrow a^+} f(x)$ means the limit as x approaches to a from the right/positive side.

#Continuity of a Function

A function $f(x)$ is continuous at $x = a$ if the left hand limit, right hand limit and functional value are simultaneously equal at point. i.e.

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

or,

$$\lim_{x \rightarrow a} f(x) = f(a)$$

#Discontinuity of a Function and Its Types

A function $f(x)$ is discontinuous at $x = a$ if the left hand limit, right hand limit and functional value are not simultaneously equal at that point. Discontinuity may be of the following types:

Jump/Ordinary discontinuity

If $\lim_{x \rightarrow a} f(x)$ does not exist i.e. $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$ then the function $f(x)$ is said to have jump/ordinary discontinuity at $x = a$.

Removable discontinuity

If $\lim_{x \rightarrow a} f(x) \neq f(a)$ provided that the limit exists i.e. $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$ then the function $f(x)$ is said to have removable discontinuity at $x = a$.

Infinite discontinuity

If $\lim_{x \rightarrow a} f(x) \rightarrow \pm\infty$ then the function $f(x)$ is said to have infinite discontinuity at $x = a$.